

Table 3: The future of AI

Aspect	Key points
Smarter, more context-aware AI	AI will become more adept at understanding project context and business goals, leading to better code suggestions.
AI-powered collaboration	AI will assist in real-time collaboration, improving team productivity and speeding up development cycles.
Democratising software development	AI tools will become more accessible, enabling non-technical users to create software and fostering innovation.
Ethical and regulatory developments	Ethical frameworks and regulations will be established to ensure responsible AI use, addressing issues like bias and privacy.
Continued integration with other technologies	AI will integrate with emerging technologies like blockchain and quantum computing, simplifying complex tasks and expanding possibilities.

functional solutions that are totally inefficient or suboptimal with regard to a specific project.

The risk here is that developers, particularly junior developers, may develop an unhealthy reliance on AI-generated suggestions without grasping the underlying logic or potential code shortcomings. AI could then inadvertently propel performance issues, bugs, or security vulnerabilities. As developers are responsible for ensuring that the code is robust, secure, and efficient, any AI-generated code should be subjected to the same rigorous review as code produced manually by a human.

Intellectual property and licensing: The other big issue is with respect to intellectual property (IP) and licensing. Generative artificial intelligence tools depend on massive datasets, which are public repositories of code with specific licences like the GNU General Public License (GPL), Apache License, etc. Thus, whenever AI generates code based on this data, there is a high chance that certain parts may be licensed, leading to possible copyright infringement.

For instance, in case the AI-generated code reflects very closely some function from a GPLed project, it could call for a violation of the agreement of that licence. Developers have to be very cautious about generating a portion of code for commercial products using AI tools,

keeping in mind relevant licences and IP laws. Tools that provide attributions or checks for licences, like automated legal scanning, can help mitigate the risks.

Bias that comes with data quality: Generative AI models are trained on large datasets that include outdated or biased coding practices. If the data used for training is either limited or lacks a modern edge, AI may repeat those defects and produce biased or exclusionary code. For example, outdated solutions could be suggested, or AI could forget factoring in the needs of inclusivity and access. Use of rich, ethical, and modern datasets must be ensured when training AI models.

The balance between human and artificial intelligence: A key challenge is finding the right balance between human intelligence and AI intelligence. If AI is overused, the problem-solving powers of developers run the risk of dwindling. If AI codes more than developers, they may not attain the level of deep clarity that is a must in critical determination-making. The goal should be not to replace developers but rather help them to work smarter as well as control and direct software development.

The future

AI will evolve as a real-time software development collaborator that assists in brainstorming, debugging, and even project management. With this collaboration, we can develop in shorter cycles, enhance team productivity, and create innovative solutions.

Generative AI will also usher in a radical change in the accessibility of software development for people who may not have a technical background. As these tools become easier to operate, many non-technical people will develop software, creating a wave of innovation and a diverse tech ecosystem.

However, ethical and regulatory frameworks will be critical as AI becomes rampant. Governments and industry bodies will come together to create responsible AI frameworks to handle concerns like bias, privacy, and job displacement.

AI is increasingly becoming context-sensitive. It is empowering developers to work smarter, effectively, and inclusively towards democratisation in software development, even making it possible for a layperson to create applications independently. Nevertheless, ethical and regulatory frameworks will remain paramount in ensuring its success. **END** 

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